

RELAPSING FEVERS

- A group of acute infections caused by arthropod born spirochetes of the genus *Borrelia*.
- *Characterized by recurrent cycles of febrile episodes, separated by asymptomatic intervals of apparent recovery.*

ETIOLOGY

- A single organism, ***Borrelia recurrentis***, is the cause of louse borne relapsing fever.
- Several different ***Borrelia*** species cause tick born relapsing fever.
- In Africa ***Borreli duttoni***, and ***Borrelia croicuidare*** are the predominant species.
- ***Borrelia*** are slender actively motile spirochetes. and measure 10-20μ long and 0.2-0.5μ wide, with 4-10 loose coils.

ETIOLOGY cont...

- Under electron microscope, 9-11 flagella are observed.
- They readily stain with aniline dyes e.g. Wright stain.
- Several species of tick borne ***Borrelia***, and recently ***B.recurrentis*** had been cultivated and propagated in artificial media.

EPIDEMIOLOGY

- **There are two epidemiological forms:**
 - Louse borne relapsing fever
 - Tick borne relapsing fever
- **Louse borne relapsing fever is transmitted:**
 - **ONLY** between humans, by the body louse, *Pediculus humanus, var corporis*.
 - It is endemic in Ethiopia, the Sudan, and Rwanda.
 - It is a disease of poverty, overcrowding, poor personal hygiene, and infestation with lice.

EPIDEMIOLOGY Cont...

- Tick borne relapsing fever is a zoonosis, maintained in nature between ticks and its natural host, often wild rodents.
- Several species of soft bodied ticks, genus, *Ornithodoros*, transmit tick born relapsing fever.
- Ticks remain infected for life, and can transmit the infection to their offspring.

PATHOGENESIS

- Portal of entry, infected lice crushed into abraded skin.
- Incubation period, 5-10 days.
- High level spirochetemia.
- Patients' producing neutralizing antibodies, clearing of the circulating strain ***Borrelia*** in 3-5 days
- New **ANTIGENIC VARIANTS** appear
 - Recurrence of clinical symptom/signs;
 - up to 3-5 relapses may occur

CLINICAL FEATURES

- Incubation period, 5-10 days, average 7 days.
- The range of clinical symptoms/ signs is wide.
- In a typical severe case there is abrupt onset of fever, 39-40.
- The following is the dominant clinical features:

CLINICAL FEATURES, contd.

SYMPTOMS

- Fever
- Headaches
- Arthralgia/myalgia
- Dry cough
- Epistaxis/gum bleeding

SIGNS

- Temperature
- Tachycardia
- Hepatomegaly
- Splenomegaly
- Petechiae/
Subconjunctival bleeding
- Jaundice
- Confusion/Meningism

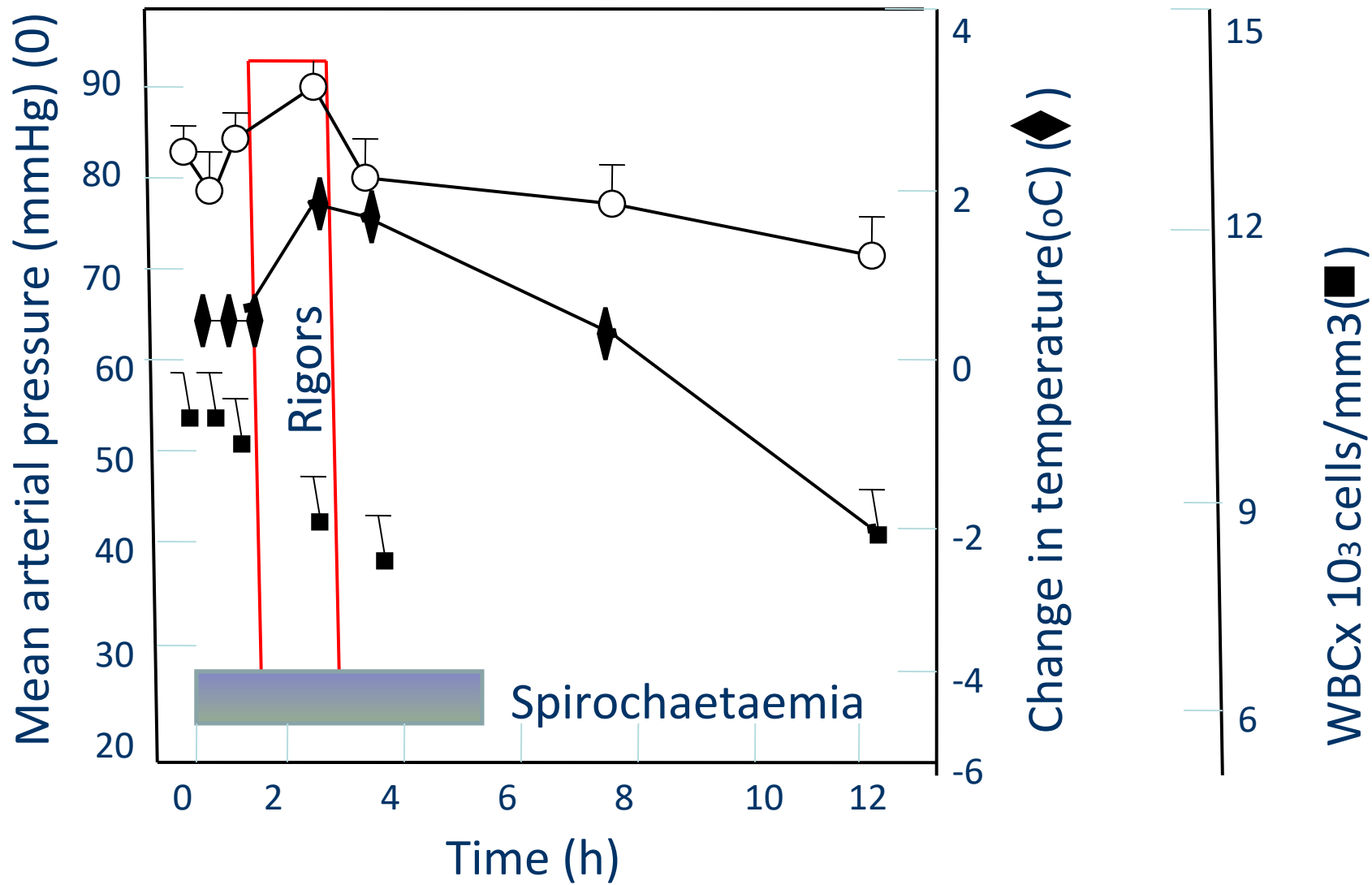


Figure-1

Fold increase over admission

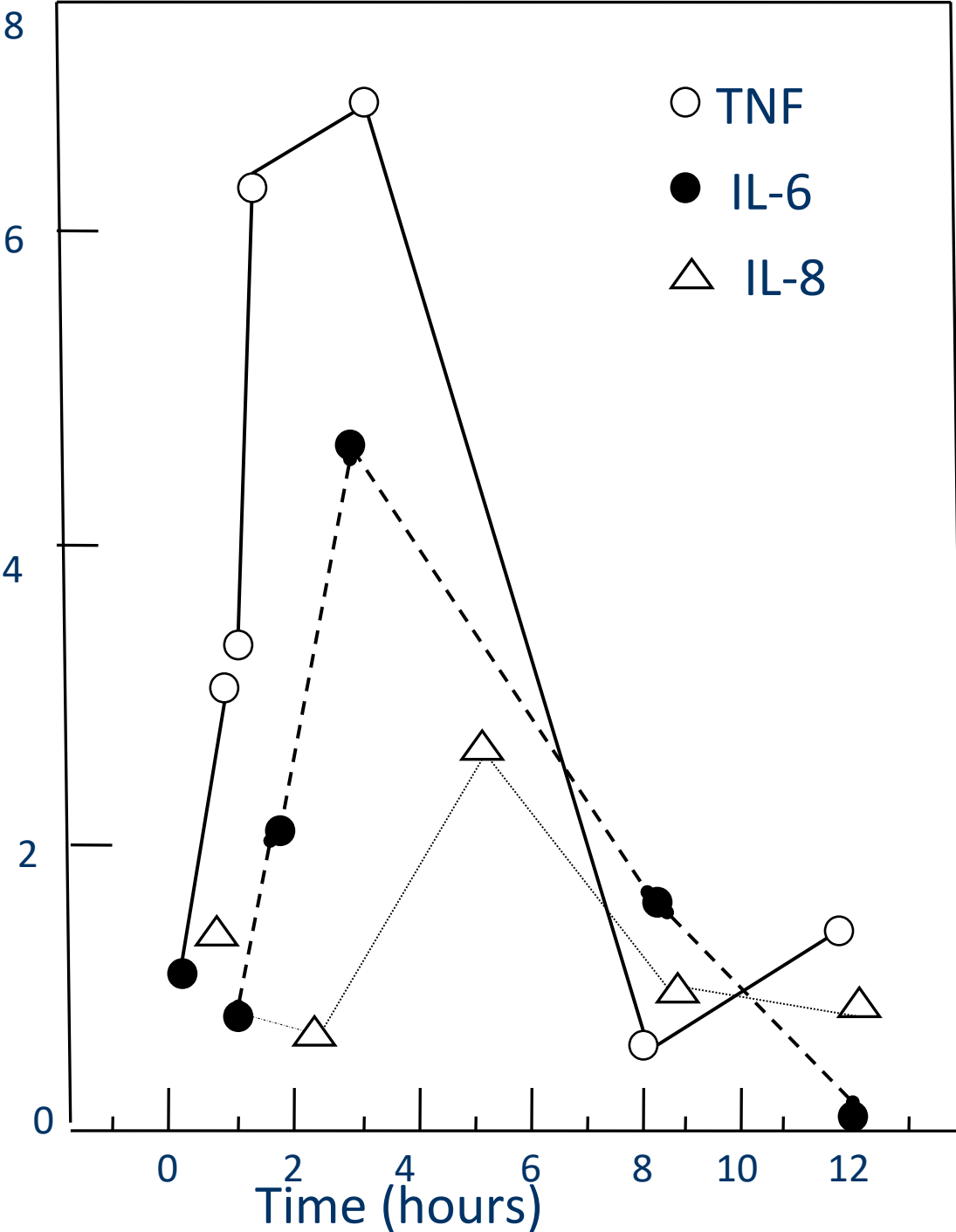


Figure-1 . Clinical characteristics of the Jarisch-Herxheimer reaction. At the indicated time points after administration of penicillin to patients with blood- smear

proven relapsing fever, venous blood samples were drawn and cytokine levels were measured in plasma. Each value is the mean $\pm$ SEM for 6-14 patients at each time point. Values on the second line represent the range observed at each time point WBC, white blood cells.

COMPLICATIONS

- Congestive heart failure
- Jaundice
- Bleeding diathesis
- Jarish- Herxheimer reaction:
 - The first dose of appropriate antibiotic causes transient worsening of clinical symptoms/signs.
 - Frequency, 35-100%,
 - associated with increased mortality.

COMPLICATIONS contd.

- **Physiologic change:**

- chills phase,
 - rise in BP, pulse, and respiratory rate;
- flush phase,
 - BP falls dramatically (figure 1).

- **There is a marked, but transient rise in circulating level of :**

- TNF, IL-6, and IL-8 at the peak of the reaction (figure 2).

DIAGNOSIS

- **Demonstration of the organism; blood film.**

MANAGEMENT

Aims of Management:

- **Clinical cure**
- **Prevention of relapse**
- **Prevention/treatment of complications**

Antibiotic treatment:

A number of antibiotics are effective:

- Anticipate the occurrence of Jarish- Herxheimer like reaction.
- In our setting we simply start with procaine penicillin 400,000IU IM and then observe for 12 hours and repeat blood film.

Management of complications

- Supportive treatment:
 - IV fluids for hypotension,
 - diuretics for pulmonary edema.

Prevention and control

- Better housing,
- reliable water supply,
- good personal hygiene.
- Insecticides like DDT for killing lice.

Rickettsial Diseases(Typhus)

- Rickettsial diseases are acute febrile illnesses caused by bacterial of the genus ***Rickettsia***.
- They are transmitted to humans by arthropod vectors such as, lice, fleas, ticks, and mites.

There are several clinical and epidemiological forms, the major forms are:

Disease	Causative agent	Geographic distribution	Arthropod vector	Natural host
<u>Typhus fever group</u> •Epidemic typhus & Brill Zinsser disease •Endemic typhus	R. Prowazekii R. typhi	World wide World wide	Body lice Flea	Man Small rodents
<u>Spotted fever group</u> •Rocky Mountain Spotted fever •Mediterranean spotted fever	R.ricketssii R.conorii	Western hemisphere Africa, Middle east, Europe	Ticks Ticks	Wild rodents, dogs Wild rodents, dogs

Forms Contd...

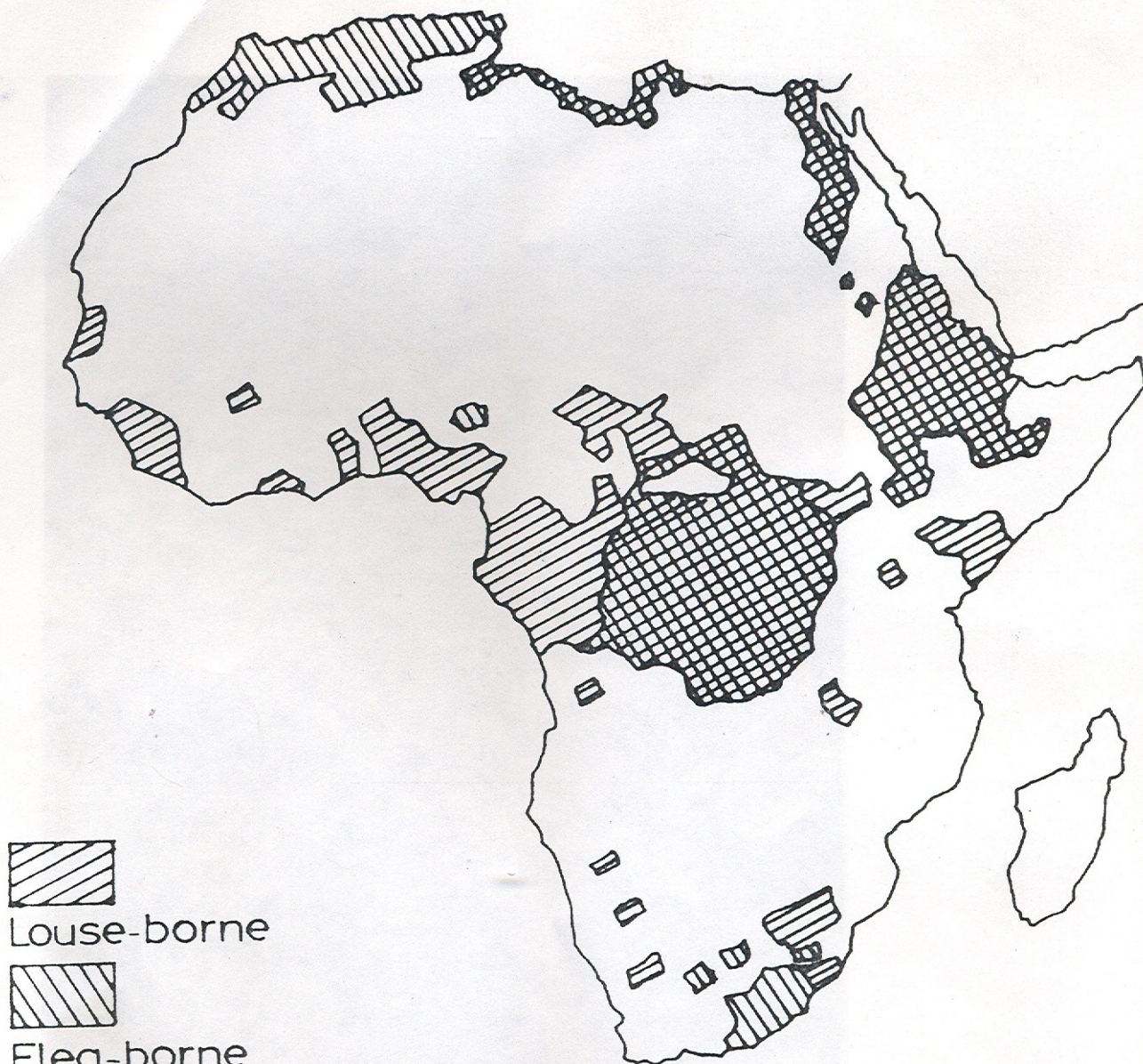
Disease	Causative agent	Geographic distribution	Arthropod vector	Natural host
<u>Others,</u> • Q fever • Trench fever	Coxiella burnetii Bartonella quintana*	World wide World wide	Ticks Body lice	Small Mammals Man

Louse born (epidemic) typhus epidemiology

- Most rickettsial diseases are zoonoses,
 - maintained in nature by a cycle between arthropod vector and mammalian host
- Man is the **ONLY** reservoir in louse borne typhus, and trench fever
 - *R. Prowazekii* had been isolated from flying squirrels in N.America; its epidemiological significance not clear

Louse borne Typhus contd.

- Louse born typhus is endemic in the highlands of Ethiopia, and it also occurs in sporadic epidemics.
- Louse borne typhus (epidemic typhus), is a classic disease of poverty, overcrowding, and infestation with lice.
- Persons at risk, prisoners, soldiers, homeless, medical personnel.



PATHOGENESIS

- Transmission is through contamination of skin abrasions/bit wound by infected lice feces.
- The infective dose, ID50, is less than 10 organisms.
- Rickettsia multiply inside endothelial cells of small blood vessels in skin, brain, liver heart and kidneys.

CLINICAL FEATURES

- Incubation period, seven days.
- Prodromal symptoms, malaise, myalgia, headache.
- Abrupt onset of fever and chills.
- Rash appears after 4-5 days.
- The rash is macular/papular/petechial.

COMPLICATIONS

- **Neurological complications, delirium, coma.**
- **Macro vascular injuries, mainly gangrene, hemiplegia.**
- **Brill-Zinsser disease, recrudescent typhus,**
 - Recurrence of fever several months/years after apparent recovery from epidemic typhus.
 - Clinical manifestation are similar to epidemic typhus, but usually milder.

DIAGNOSIS

- A compatible clinical illness, with positive serological test
- A four- fold rise in titer, or single titer $>1/160$

SEROLOGICAL REACTIONS IN RICKETTSIAL INFECTIONS

Type of ricketssial infection	PROTEUS*		CF	MA	IFA
	OX19	OXK			
Louse born (Epidemic), Typus	+	0	+	+	+
Flea Born (endemic) Typhus	+	0	+	+	+
Rocky Mountain Spotted fever	+	0	+	+	+
Scrub typhus	0	+	+	+	+

*Weil-Felix Test

CF, complement fixation

MA, macroscopic agglutination

IFA, indirect fluorescent antibody

MANAGEMENT

- Several antibiotics are effective:
 - tetracycline,
 - doxycycline, and
 - chloramphenicol
- Shorter courses/single dose are as effective as traditionally recommended 7-10 days of treatment.